

Problem 6: Semantic Atlas

(13 marks)

Have you ever wondered about how language might be represented in our brain? Recent work done at the University of California, Berkeley found that words with similar meanings or connotations seem to be represented in similar regions of the brain (called “voxels” which are like two-dimensional “pixels” but in three dimensions), in the form of high activity in those regions when those words are processed. Given below are some words some brain activity patterns in five distinct voxels.

Word	Voxel 1	Voxel 2	Voxel 3	Voxel 4	Voxel 5
(A)	high	high	high	low	low
(B)	low	high	low	high	low
(C)	high	high	low	high	low
(D)	low	high	high	high	high
(E)	high	low	low	low	low
(F)	high	low	low	high	low
(G)	low	high	low	low	high
(H)	low	low	low	low	high
(I)	low	low	high	low	high
(J)	high	low	high	low	low
insult	low	high	high	low	high
illusion	(1)	(2)	low	(3)	(4)
murder	low	(5)	(6)	low	(7)
blood	high	high	(8)	(9)	(10)
helpful	(11)	(12)	high	(13)	(14)
belief	(15)	(16)	low	(17)	(18)

Assignment 1: (8 marks)

Match words from the following list to the appropriate spaces labelled (A)–(J) above:

fracture, stripes, depression, philosophy, torture, cut (wound/injury),
eyes, love, war film, telephone

Assignment 2: (5 marks)

Fill in the spaces numbered (1)–(18) above.

– Abhishek Dedhe